



NORTH SYDNEY BOYS HIGH SCHOOL

EXTENSION 1 MATHEMATICS

2016 HSC Course Assessment Task 2

16th March 2016

General Instructions

- Working time – 1 hours (plus 5 minutes reading time).
- Use a blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.

Section I

- Mark answers on the answer grid provided.

Section II

- Commence each question on a new page.
- Write on both sides of the paper.
- Show all necessary working for every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Teacher (please ✓) ☐ Mr Lin ☐ Mr Jooma ☐ Mr Ireland
☐ Mr Berry ☐ Ms Ziaziaris ☐ Ms Lee

Question	1-5	6	7	8	9	10	11	Total	%
Marks	$\overline{5}$	$\overline{7}$	$\overline{7}$	$\overline{5}$	$\overline{7}$	$\overline{7}$	$\overline{8}$	$\overline{46}$	$\overline{100}$

Section I

5 Marks

Attempt Questions 1 to 5

Mark your answers on the answer sheet provided on page 2.

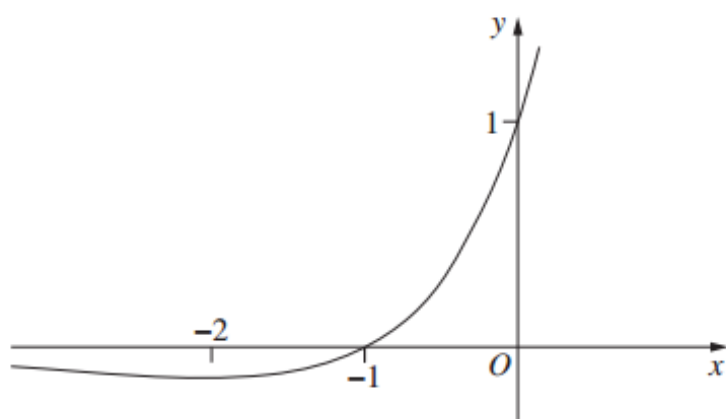
1. Find the value of $\log_4 32$
 - A) 1.5
 - B) 2
 - C) 2.5
 - D) 3

2. The solution to $\log_3(x - 1) = 3$ is?
 - A) 7
 - B) 10
 - C) 26
 - D) 28

3. Evaluate $\int_e^{e^2} \frac{2}{x} dx$
 - A) 2
 - B) 4
 - C) 6
 - D) 8

4. The derivative of $y = x \tan 2x$ is
 - (A) $\tan 2x + x \sec^2 2x$
 - (B) $\tan 2x + 2 \sec^2 2x$
 - (C) $x \tan 2x + \sec^2 2x$
 - (D) $2 \tan 2x + x \sec^2 2x$

5. The diagram shows the graph of $y = e^x(1 + x)$



How many solutions are there to the equation $e^x(1 + x) = 1 - x^2$?

- (A) 0
(B) 1
(C) 2
(D) 3

END OF SECTION I

1

A

B

C

D

2

A

B

C

D

3

A

B

C

D

4

A

B

C

D

5

A

B

C

D

Section II

Total of **41** marks

Attempt Questions 6 to 11.

Write your answers in the writing book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (7 marks) Use a **NEW** page.

Differentiate the following

(a) $y = e^{\sin x}$ **1**

(b) $y = \cos(\log x)$ **2**

(c) $y = e^{x^2} e^x$ **2**

(d) $y = \log\left(\frac{x}{x^2 - 1}\right)$ **2**

Question 7 (7 marks) Use a **NEW** page.

(a) Find the following integrals

(i) $\int \frac{2x + 3}{x^2 + 3x + 5} dx$ **1**

(ii) $\int \frac{x^2 + 2x - 3}{x^2} dx$ **2**

(b) Evaluate the following integrals

(i) $\int_0^1 x e^{x^2+3} dx$ **2**

(ii) $\int_0^1 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$ **2**

Question 8 (5 marks) Use a **NEW** page.

- (a) The area of a sector of a circle of radius 6cm is 50cm^2 . Find the length of the arc. 2
- (b) Find the equation of the tangent to the curve $y = 2 \sin(2x + \frac{\pi}{3})$ at the point where $x = \frac{\pi}{2}$. 3

Question 9 (7 marks) Use a **NEW** page.

- (a) Find the volume of the solid generated when the curve $y = \tan x$, the x axis and $x = \frac{\pi}{3}$ is rotated about the x axis. 3
- (b) Let $f(x) = 2 \cos 2x$
- (i) Sketch $f(x)$ for $0 \leq x \leq 2\pi$ 2
- (ii) Hence, find all values of x such that $f(x) \leq 1$, for $0 \leq x \leq 2\pi$ 2

Question 10 (7 marks) Use a **NEW** page.

- (a) (i) Show that $\frac{1}{2^2-1} + \frac{1}{3^2-1} + \dots + \frac{1}{p^2-1} = \frac{p(3p+5)}{4p(p+1)}$ by using the principle of mathematical induction 3
- (ii) Hence find 1
- $$\lim_{x \rightarrow \infty} \sum_{t=2}^x \left(\frac{1}{t^2 - 1} \right)$$
- (b) If $(ax)^{\log a} = (bx)^{\log b}$, prove that $x = \frac{1}{ab}$ 3

Question 11 (8 marks) Use a **NEW** Page.

Given the function $y = x^x$ for $x > 0$

- (a) What is the value of $\lim_{x \rightarrow 0} x^x$? 1
- (b) By using $x = e^{\log x}$ or otherwise, show that $\frac{dy}{dx} = x^x(1 + \log x)$ 2
- (c) Find the turning point(s) and determine its nature 3
- (d) Using the information from above, sketch the curve of $y = x^x$ 2

End of Examination

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Multiple Choice

1. C

2. D

3. A

4. B

5. C

Question 6

a) $y = e^{\sin x}$

$$\frac{dy}{dx} = \cos x \cdot e^{\sin x}$$

b) $y = \cos(\log x)$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{x} \cdot -\sin(\log x) \\ &= -\frac{\sin(\log x)}{x}\end{aligned}$$

(c) $y = e^{x^2} \cdot e^x$
 $= e^{x^2+x}$

$$\frac{dy}{dx} = (2x+1)e^{x^2+x}$$

(d) $y = \log \frac{x}{x^2-1}$

$$= \log x - \log(x^2-1)$$

$$\frac{dy}{dx} = \frac{1}{x} - \frac{2x}{x^2-1}$$

Question 7

a) (i) $\ln(x^2+3x+5) + C$

(ii) $\int \frac{x^2+2x-3}{x^2} dx$

$$= \int 1 - \frac{2}{x} - \frac{3}{x^2} dx$$

$$= x - 2\ln|x| + \frac{3}{x} + C$$

b) (i) $\int_0^1 x e^{x^2+3} dx$

$$= \frac{1}{2} [e^{x^2+3}]_0^1$$

$$= \frac{1}{2} e^3 [e-1]$$

(ii) $\int_0^1 \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$

$$= [\ln(e^x + e^{-x})]_0^1$$

$$= \ln\left(e + \frac{1}{e}\right) - \ln 2$$

$$= \ln\left(\frac{e^2+1}{2e}\right)$$

Question 8

$$a) A = \frac{1}{2} r^2 \theta$$

$$50 = \frac{1}{2} \cdot 36 \cdot \theta$$

$$\therefore \theta = \frac{25}{9}$$

$$\therefore l = r \theta$$

$$= 6 \times \frac{25}{9}$$

$$= \frac{150}{9} \text{ cm}$$

$$= \frac{50}{3} \text{ cm}$$

$$b) y = \sin\left(x + \frac{\pi}{3}\right)$$

$$\frac{dy}{dx} = \cos\left(x + \frac{\pi}{3}\right)$$

$$\text{when } x = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$m = \cos\left\{\frac{5\pi}{6}\right\}$$

$$= -\frac{\sqrt{3}}{2}$$

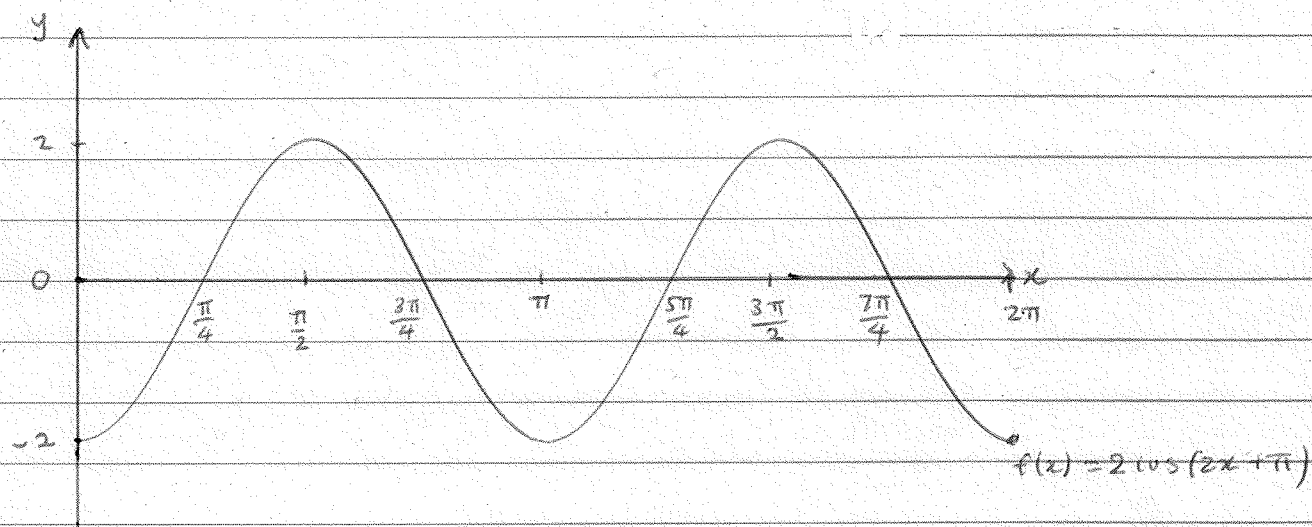
$$y - \frac{1}{2} = -\frac{\sqrt{3}}{2} \left(x - \frac{\pi}{2}\right)$$

$$-2y + 1 = \sqrt{3}x + \frac{\sqrt{3}}{2}$$

$$2\sqrt{3}x + 4y + (\sqrt{3} - 2) = 0$$

Question 9

$$\begin{aligned} \text{a) } f(x) &= 2 \cos(2x + \pi) \\ &= -2 \cos(2x) \end{aligned}$$



$$\text{b) } (ax)^{\log a} = (bx)^{\log b}$$

$$\log(ax)^{\log a} = \log(bx)^{\log b}$$

$$\log a \cdot \log(ax) = \log b \cdot \log(bx)$$

$$\log a (\log a + \log x) = \log b (\log b + \log x)$$

$$(\log a)^2 + \log a \cdot \log x = (\log b)^2 + \log b \cdot \log x$$

$$(\log a)^2 - (\log b)^2 = (\log b - \log a) \cdot \log x$$

$$(\log a - \log b)(\log a + \log b) = \log x \cdot (\log b - \log a)$$

$$\log(ab) = -\log x$$

$$\log x = \log\left(\frac{1}{ab}\right)$$

$$x = \frac{1}{ab}$$

Question 10

Prove true for the base case. $p=2$.

$$\text{LHS} = \frac{1}{3} \quad \text{RHS} = \frac{(2-1)(6+2)}{8(2+1)} \\ = \frac{1}{3}$$

$\therefore$ true for $p=2$

Assume true for $p=k$ where $k \geq 2, k \in \mathbb{Z}$.

$$\text{i.e. } \frac{1}{2^2-1} + \frac{1}{3^2-1} + \dots + \frac{1}{k^2-1} = \frac{(k-1)(3k+2)}{4k(k+1)} \quad (*)$$

Required to prove true for $p=k+1$

$$\text{i.e. } \frac{1}{2^2-1} + \frac{1}{3^2-1} + \dots + \frac{1}{k^2-1} + \frac{1}{(k+1)^2-1} = \frac{k(3k+5)}{4(k+1)(k+2)}$$

$$\begin{aligned} \text{Now LHS} &= \frac{(k-1)(3k+2)}{4k(k+1)} + \frac{1}{(k+1)^2-1} \quad \text{from the induction hypothesis } (*) \\ &= \frac{(k-1)(3k+2)}{4k(k+1)} + \frac{1}{k(k+2)} \\ &= \frac{(k-1)(3k+2)(k+2) + 4(k+1)}{4k(k+1)(k+2)} \\ &= \frac{3k^3 + 5k^2}{4k(k+1)(k+2)} \\ &= \frac{k(3k+5)}{4(k+1)(k+2)} \\ &= \text{RHS} \end{aligned}$$

$\therefore$ true for $p=k+1$

$\therefore$ Since it's true for $p=2$ and $p=k+1$, then it is true for all integers $p \geq 2$ by the principle of mathematical induction.

Question 11

a) $\lim_{x \rightarrow 0} x^x = 1$

b) $y = x^x = e^{x \log_e x}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (x \log_e x) \cdot e^{x \log_e x} \\ &= \left(x \cdot \frac{1}{x} + \log_e x\right) \cdot e^{x \log_e x} \\ &= (1 + \log_e x) \cdot x^x\end{aligned}$$

c) stationary point occurs when $\frac{dy}{dx} = 0 \Rightarrow (1 + \log x) x^x = 0$
since $x^x \neq 0$ for all values.

$$1 + \log x = 0$$

$$x = \frac{1}{e} \approx 0.368$$

$$y = \left(\frac{1}{e}\right)^{\frac{1}{e}} \approx 0.692$$

x	0.3	$\frac{1}{e}$	0.4
$\frac{dy}{dx}$	-0.14	0	0.06

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$\therefore$ minimum turning point at $\left(\frac{1}{e}, \left(\frac{1}{e}\right)^{\frac{1}{e}}\right)$

d)

